

Challenge (Habanero)

- Q1.** A recipe for cooking pasta requires $2 \cdot x$ cups of water for every x pounds of pasta. What is the rate of change of water with respect to pasta?
(A) 2
(B) 1
(C) 0.5
(D) 4
(E) 0.25
- Q2.** The height of a building is $5 \cdot t + 20$ feet after t hours of construction. What is the rate of change of the building's height?
(A) 20 feet/hour
(B) 5 feet/hour
(C) 10 feet/hour
(D) 2 feet/hour
(E) 1 foot/hour
- Q3.** The cost of ingredients for a cake is $15 \cdot x + 10$ dollars for x cakes. What is the slope of the cost function?
(A) 15
(B) 10
(C) 25
(D) 5
(E) 20
- Q4.** A contractor is building a road with a slope of 3. If the road is x miles long, how much higher is the end of the road than the beginning?
(A) $3x$ feet
(B) $3x$ miles
(C) 3 miles
(D) x miles
(E) $3x$ yards
- Q5.** The temperature in a kitchen is $2 \cdot t + 70$ degrees after t hours of cooking. What is the rate of change of the temperature?
(A) 70 degrees/hour
(B) 2 degrees/hour
(C) 1 degree/hour
(D) 0.5 degrees/hour
(E) 5 degrees/hour
- Q6.** A bridge is 50 feet long and 10 feet high at one end. If the bridge has a uniform slope, what is the slope of the bridge?
(A) 0.2
(B) 0.5
(C) 1
(D) 2
(E) 5
- Q7.** A water tank is being filled at a rate of 5 gallons per minute. If the tank is x gallons full after t minutes, what is the equation of the tank's volume?
(A) $x = 5t$
(B) $x = 5 \cdot t + 10$
(C) $x = 10t$
(D) $x = 10 \cdot t + 5$
(E) $x = 2t$
- Q8.** The cost of fuel for a construction vehicle is $0.5 \cdot x + 20$ dollars for x hours of operation. What is the slope of the cost function?
(A) 0.5
(B) 1
(C) 2
(D) 5
(E) 10

Open Ended Questions (FRQ)

- Q9.** A construction company is building a ramp with a uniform slope. The ramp is 20 feet long and 5 feet high at one end. Find the slope of the ramp and explain what it represents in the context of the problem.
- Q10.** A recipe for making bread requires $2 \cdot x$ cups of flour for every x loaves of bread. If the cost of flour is 1.50 dollars per cup, find the total cost of flour for x loaves of bread and explain how the rate of change of the cost relates to the number of loaves.

Chef's Tips

- Read each question carefully before answering.
- Use a ruler to draw graphs and diagrams.
- Show all work and explain your reasoning.